

1.)

$$\begin{aligned} \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) &= \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2) = \frac{1}{4} (\langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle) = \\ &= \frac{1}{4} (\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle - \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle) = \\ &= \frac{1}{4} (\langle u, v \rangle + \langle u, v \rangle + \langle u, v \rangle + \langle v, u \rangle) = \frac{1}{4} \cdot 2 (\langle u, v \rangle + \overline{\langle u, v \rangle}) = \\ &= \frac{1}{2} (2 \operatorname{Re}(\langle u, v \rangle)) = \underline{\operatorname{Re}(\langle u, v \rangle)} \quad \checkmark \end{aligned}$$

$x + \bar{x} = 2 \operatorname{Re}(x)$

~~Do~~ Dôkaz je iba rozpisanie prvej strany za použitia operácií definovaných v obecnom sk. 5.

$$\begin{aligned} \operatorname{Im}(\langle u, v \rangle) &= \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2) = \frac{1}{4} (\langle u - iv, u - iv \rangle - \langle u + iv, u + iv \rangle) = \\ &= \frac{1}{4} (\langle u, u \rangle - \langle u, iv \rangle - \langle iv, u \rangle + \langle iv, iv \rangle - \langle u, u \rangle - \langle u, iv \rangle - \langle iv, u \rangle - \langle iv, iv \rangle) = \\ &= \frac{1}{4} \cdot 2 (-\langle u, iv \rangle - \langle iv, u \rangle) = \frac{1}{4} \cdot 2 (-i\langle u, v \rangle + i\langle v, u \rangle) = \\ &= \frac{1}{2} (-i\langle u, v \rangle + i\overline{\langle u, v \rangle}) = \frac{1}{2} (2 \operatorname{Im}(\langle u, v \rangle)) = \underline{\operatorname{Im}(\langle u, v \rangle)} \quad \checkmark \end{aligned}$$

$$-iz + i\bar{z} = \operatorname{Im}(z)$$

$$z = a + ib$$